

## SAFETY DATA SHEET

Creation Date 21-January-2011

Revision Date 29-March-2024

Revision Number 5

### 1. Identification

**Product Name** Potassium cyanide

**Cat No. :** 12136

**CAS-No** 151-50-8  
**Synonyms** Cyanide of potassium; Hydrocyanic acid, potassium salt; KCN.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
 112 Colonnade Road,  
 Ottawa, ON K2E 7L6,  
 Canada  
 Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |            |
|-------------------------------------------------------------|------------|
| <b>Corrosive to metals</b>                                  | Category 1 |
| <b>Acute oral toxicity</b>                                  | Category 1 |
| <b>Acute dermal toxicity</b>                                | Category 1 |
| <b>Acute Inhalation Toxicity</b>                            | Category 1 |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 1 |
| Target Organs - Heart, Cardiovascular system.               |            |
| <b>Health Hazards Not Otherwise Classified</b>              | Category 1 |
| Contact with acids liberates very toxic gas                 |            |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

May be corrosive to metals

Fatal if swallowed, in contact with skin or if inhaled

Causes damage to organs through prolonged or repeated exposure

Contact with acids liberates very toxic gas

**Precautionary Statements****Prevention**

Take any precaution to avoid mixing with acids

Do not breathe dust/fumes/gas/mist/vapours/spray

Wear respiratory protection

Keep only in original container

Do not get in eyes, on skin, or on clothing

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing

Immediately call a POISON CENTER/doctor

IF ON SKIN: Wash with plenty of soap and water

Rinse mouth

Absorb spillage to prevent material damage

Take off immediately all contaminated clothing and wash it before reuse

**Storage**

Store locked up

Store in a well-ventilated place. Keep container tightly closed

Store in corrosive resistant polypropylene container with a resistant inliner

Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component         | CAS-No   | Weight % |
|-------------------|----------|----------|
| Potassium cyanide | 151-50-8 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

|                                        |                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                           |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.                        |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                        |
| <b>Most important symptoms/effects</b> | Systemic Toxicity: Respiratory disorders: Symptoms may include tightness in the chest, flushing, headache, nausea, vomiting, respiratory depression, weakness, irregular heartbeat, abdominal pain, convulsions, and shock: May cause cyanosis (bluish discoloration of skin due to deficient oxygenation of the blood): Exposure may result in death |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                 |

## 5. Fire-fighting measures

|                                         |                                                                                                         |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                |
| <b>Flash Point</b>                      | No information available                                                                                |
| <b>Method -</b>                         | No information available                                                                                |
| <b>Autoignition Temperature</b>         | Not applicable                                                                                          |
| <b>Explosion Limits</b>                 |                                                                                                         |
| <b>Upper</b>                            | No data available                                                                                       |
| <b>Lower</b>                            | No data available                                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                |

### Specific Hazards Arising from the Chemical

Non-combustible. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Nitrogen oxides (NOx). Hydrogen cyanide (hydrocyanic acid). Potassium oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 4             | 0                   | 1                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                        |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation.                                                                                                    |

## 7. Handling and storage

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face |
|-----------------|--------------------------------------------------------------------------------------|

protection. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep locked up. Keep away from acids. Keep away from combustible material. Do not store in aluminum containers. Incompatible Materials. Acids. Strong oxidizing agents. Bases. Finely powdered metals. Aldehydes. Peroxides. Metals.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component         | Alberta                              | British Columbia                     | Ontario TWAEV                    | Quebec                                                   | ACGIH TLV                            | OSHA PEL                           | NIOSH                                                                          |
|-------------------|--------------------------------------|--------------------------------------|----------------------------------|----------------------------------------------------------|--------------------------------------|------------------------------------|--------------------------------------------------------------------------------|
| Potassium cyanide | Ceiling: 5 mg/m <sup>3</sup><br>Skin | Ceiling: 5 mg/m <sup>3</sup><br>Skin | CEV: 5 mg/m <sup>3</sup><br>Skin | Ceiling: 10 ppm<br>Ceiling: 11 mg/m <sup>3</sup><br>Skin | Ceiling: 5 mg/m <sup>3</sup><br>Skin | (Vacated) TWA: 5 mg/m <sup>3</sup> | IDLH: 25 mg/m <sup>3</sup><br>Ceiling: 4.7 ppm<br>Ceiling: 5 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments                                                                    |
|----------------|-------------------|-----------------|-----------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.35 mm         | As tested under EN374-3<br>Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                            |
|----------------------------------------|----------------------------|
| Physical State                         | Powder Solid               |
| Appearance                             | White                      |
| Odor                                   | bitter almonds             |
| Odor Threshold                         | No information available   |
| pH                                     | 11-12 20 g/l aq.sol.(20°C) |
| Melting Point/Range                    | 634 °C / 1173.2 °F         |
| Boiling Point/Range                    | 1625 °C / 2957 °F          |
| Flash Point                            | No information available   |
| Evaporation Rate                       | Not applicable             |
| Flammability (solid,gas)               | No information available   |
| Flammability or explosive limits       |                            |
| Upper                                  | No data available          |
| Lower                                  | No data available          |
| Vapor Pressure                         | No information available   |
| Vapor Density                          | Not applicable             |
| Specific Gravity                       | 1.52 @ 16°C                |
| Solubility                             | Soluble in water           |
| Partition coefficient; n-octanol/water | No data available          |
| Autoignition Temperature               | Not applicable             |
| Decomposition Temperature              | No information available   |
| Viscosity                              | Not applicable             |
| Molecular Formula                      | C K N                      |
| Molecular Weight                       | 65.12                      |

## 10. Stability and reactivity

|                                  |                                                                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                                 |
| Stability                        | Moisture sensitive.                                                                                                                                 |
| Conditions to Avoid              | Excess heat. Burning produces obnoxious and toxic fumes. Incompatible products. Exposure to light. Exposure to moist air or water. Exposure to air. |
| Incompatible Materials           | Acids, Strong oxidizing agents, Bases, Finely powdered metals, Aldehydes, Peroxides, Metals                                                         |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Hydrogen cyanide (hydrocyanic acid), Potassium oxides                                                                        |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                            |
| Hazardous Reactions              | None under normal processing.                                                                                                                       |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component         | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation              |
|-------------------|---------------------------|------------------------------|------------------------------|
| Potassium cyanide | LD50 = 7.49 mg/kg ( Rat ) | LD50 = 22.3 mg/kg ( Rabbit ) | LC50 = 0.16 mg/L ( Rat ) 1 h |

Toxicologically Synergistic Products No information available

#### Products

Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Irritating to eyes, respiratory system and skin                                          |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component         | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------|----------|------------|------------|------------|------------|------------|
| Potassium cyanide | 151-50-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** Heart Cardiovascular system

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Systemic Toxicity: Respiratory disorders: Symptoms may include tightness in the chest, flushing, headache, nausea, vomiting, respiratory depression, weakness, irregular heartbeat, abdominal pain, convulsions, and shock: May cause cyanosis (bluish discoloration of skin due to deficient oxygenation of the blood): Exposure may result in death

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component         | Freshwater Algae | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                        | Microtox   | Water Flea |
|-------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|
| Potassium cyanide | Not listed       | LC50: 0.31 - 0.37 mg/L, 96h static (Pimephales promelas)<br>LC50: = 0.0588 mg/L, 96h flow-through (Poecilia reticulata)<br>LC50: 0.45 - 0.57 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: = 0.45 mg/L, 96h (Lepomis macrochirus)<br>LC50: 0.01 - 0.08 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 0.044 - 0.084 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 0.04 - 0.046 mg/L, 96h flow-through (Oncorhynchus mykiss) | Not listed | Not listed |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                    | RCRA - U Series Wastes | RCRA - P Series Wastes  |
|------------------------------|------------------------|-------------------------|
| Potassium cyanide - 151-50-8 | -                      | not otherwise specified |

### 14. Transport information

#### DOT

UN-No UN1680  
 Proper Shipping Name POTASSIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Packing Group I

#### TDG

UN-No UN1680  
 Proper Shipping Name POTASSIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Packing Group I

#### IATA

UN-No UN1680  
 Proper Shipping Name POTASSIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Packing Group I

#### IMDG/IMO

UN-No UN1680  
 Proper Shipping Name POTASSIUM CYANIDE, SOLID  
 Hazard Class 6.1  
 Packing Group I

### 15. Regulatory information

#### International Inventories

| Component         | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Potassium cyanide | 151-50-8 | X   | -    | X    | ACTIVE                                        | 205-792-3 | -      | -   |

| Component         | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------|----------|-------|----------|------|------|------|------|-------|-------|
| Potassium cyanide | 151-50-8 | X     | KE-29092 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant | Canadian Environmental | Canada's Chemicals Management |
|-----------|-----------------------------|------------------------|-------------------------------|
|-----------|-----------------------------|------------------------|-------------------------------|

|                   | Release Inventory (NPRI)  | Protection Agency (CEPA)<br>- List of Toxic Substances | Plan (CEPA) |
|-------------------|---------------------------|--------------------------------------------------------|-------------|
| Potassium cyanide | Part 1, Group A Substance |                                                        |             |

**Legend** NPRI - National Pollutant Release Inventory

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component         | REACH (1907/2006) - Annex XIV -<br>Substances Subject to<br>Authorization | REACH (1907/2006) - Annex XVII -<br>Restrictions on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 - Candidate<br>List of Substances of Very High<br>Concern (SVHC) |
|-------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Potassium cyanide | -                                                                         | Use restricted. See item 75.<br>(see link for restriction details)                  | -                                                                                                              |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component         | CAS-No   | OECD HPV | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|-------------------|----------|----------|---------------------------------|------------------------------|--------------------------------------------------|
| Potassium cyanide | 151-50-8 | Listed   | Not applicable                  | Not applicable               | Not applicable                                   |

| Component         | CAS-No   | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|-------------------|----------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Potassium cyanide | 151-50-8 | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Annex I - Y33                         |

## 16. Other information

#### Prepared By

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

#### Creation Date

21-January-2011

#### Revision Date

29-March-2024

#### Print Date

29-March-2024

#### Revision Summary

New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**